RV College of Engineering®

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Academic Year 2024-25 (ODD Semester)

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Department of Artificial Intelligence and Machine Learning

Course Code:	:	AI254TA	Date	:	28/01/2025
Semester	:	V	Time	:	2:000-3:30
Max Marks	:	50	Duration	:	90 mins

Machine Learning Operations (MLOps)

Improvement CIE

Note: Answer all the Questions

SL. No		PART B Questions	M	BT	CO
1	a	Elaborate the different steps involved in CI/CD pipeline. Give an example for the implementation of a ML model using the pipeline	10	3	5
2	b	List any two challenges that occur during deployment of an ML Model and how does kubernets overcome the challenges? Give an example for the same.	6	2	3
	b	Consider a recommendation engine and describe how Ground Truth Evaluation can be used to verify the model degradation. What are the various metrics that are used by ground truth for the same	4	2	5
3	a	Elaborate the factors on which the data scientist decides to retrain the ML Model.	5	2	3
3	b	Explore the various information available in event log ?	5	2	5
4		A large financial institution implemented a machine learning model to detect fraudulent transactions. The model was trained on historical transaction data, including features such as transaction amount, merchant category, user location, and transaction time. Over time, the institution noticed a decline in the model's accuracy. The institution believed data drift as one of the primary reasons for model degradation (i) What are the causes for input drift detection . (ii) What are the different techniques to handle them ?	10	3	5
5	a	Considering a pharmaceutical ML model that has to be deployed in US, discuss the quality guidelines established by GxP.	5	2	5
	b	List the different elements of Responsible AI. Elaborate any two	5	2	5

M-Marks, BT-Blooms Taxonomy Levels, CO-Course Outcomes

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks Part B			11	39			30	20			

Course Outcomes: After completing the course, the students will be able to:-

CO1	Identify and apply various ML-Ops techniques to deploy machine learning models for real-world problems.
CO2	Design, deploy and evaluate Machine Learning models, follow the operational practices to benefit society, science, and industry.
CO3	Use modern tools and techniques to organize ML model from development to production for real world problems
CO4	Demonstrate effective communication through team presentations and reports to analyse the impact of the standard MLOPs practices on industry and society.
CO5	Conduct performance evaluation, design, deploy models in par with the appropriate Governance for the benefit of the industry and society.